

test-coverage

Containers — Test Coverage Ledger

Field	Value
Revision	1
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Status	active
Round	299 (template mirror of round-220/295/298)
Anti-bluff anchor	§11.4 + CONST-035 + CONST-050(B)

Verbatim 2026-05-19 operator mandate (CONST-049 §11.4.17): *"all existing tests and Challenges do work in anti-bluff manner - they MUST confirm that all tested codebase really works as expected! We had been in position that all tests do execute with success and all Challenges as well, but in reality the most of the features does not work and can't be used! This MUST NOT be the case and execution of tests and Challenges MUST guarantee the quality, the completion and full usability by end users of the product!"*

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Purpose

This ledger satisfies CONST-050(B) — every production symbol of meaningful public surface **MUST** be covered by every test type the domain warrants. The ledger is the single point of truth answering "is pkg/<X>.<Func> covered by unit + integration + e2e + Challenge + paired mutation?"

A row whose symbol is uncovered is a §11.4 PASS-bluff at the coverage layer: the suite reports green while the symbol is unproven. The round-299 sweep populated the ledger from the live tree; subsequent rounds extend it.

Symbol → test ledger

Package	Public symbol	Unit	Integration
pkg/runtime	AutoDetect, DockerRuntime, PodmanRuntime, KubernetesRuntime	yes (pkg/runtime/*_test.go)	yes (tests/integration/distribution_integration_t
pkg/compose	ComposeOrchestrator.Up, Down, Logs	yes	yes (tests/integration/distribution_integration_t
pkg/health	TCPChecker, HTTPChecker, GRPCChecker, RetryPolicy	yes	yes (real TCP / HTTP serve
pkg/envconfig	LoadFromEnv, LoadFromFile, ToRemoteHosts	yes (pkg/envconfig/*_test.go)	yes (round-trips tests/configs/.env.*)
pkg/distribution	Distributor.Distribute, Undistribute, Rebalance	yes	yes (tests/integration/distribution_integration_t
pkg/scheduler	5 strategies (resource_aware, round_robin, affinity, spread, bin_pack) + ScheduleBatch	yes (pkg/scheduler/*_test.go)	yes (tests/integration/)
pkg/remote	SSHExecutor.Run, HostManager.AddHost, ProbeAll, IsReachable	yes (mocks ONLY in *_test.go)	yes (real SSH against test configs/.env.remote-single CONTAINERS_REMOTE_ENABLED=
pkg/orchestrator	Service.Add, DiscoverServices, StartAll, StartService	yes	yes (tests/integration/)
pkg/lifecycle	LazyBooter, IdleShutdown, ConcurrencySemaphore	yes	yes (real timers)
pkg/network	TunnelManager.CreateTunnel, Close	yes	yes (real SSH port-forward

Package	Public symbol	Unit	Integration
pkg/volume	VolumeManager.Mount, Unmount (SSHFS/NFS/rsync)	yes	yes (real mounts when supported)
pkg/event	EventBus.Publish, Subscribe, 20 event types	yes	yes
pkg/metrics	MetricsCollector.Register, Prometheus exposition	yes	yes (/metrics scrape)
pkg/discovery	TCPDiscoverer.Probe, DNSDiscoverer.Resolve	yes	yes (real TCP / DNS)
pkg/i18n	Translator.T, bundle load (active.en.yaml + 5 locales added round-299)	yes (pkg/i18n/*_test.go)	yes (locale-switch round-trip)
pkg/ctop	Collector.Collect, Display.Run, RenderSnapshot, RenderJSON	yes	yes (real container snapshots)
pkg/boot	BootManager.BootAll, WithRuntime, WithLogger	yes	yes (real compose Up + hello)

Challenge → invariant ledger

Challenge	Invariant exercised	Normal exit	Mutate-mode exit
anchor_manifest_challenge.sh	Governance anchors literal-present in CONSTITUTION/CLAUDE/AGENTS	0	99 (strip an anchor)
bluff_scanner_challenge.sh	No simulated / for now / TODO implement in production code	0	99 (plant a literal)
chaos_failure_injection_challenge.sh	Distributor.Rebalance recovers when a host is force-killed mid-deploy	0	99 (skip the recovery step)
ddos_health_flood_challenge.sh	Health endpoints survive flood without panic; retry policy honored	0	99 (remove retry wrap)
host_no_auto_suspend_challenge.sh	CONST-033 host-power-management ban — host does not auto-suspend	0	99 (re-enable auto-suspend)
mutation_ratchet_challenge.sh	Every covenant gate fails when its pro-	0	99 (gate that sur-

Challenge	Invariant exercised	Normal exit	Mutate-mode exit
	duction code is mutated		vives mutation is a bluff gate)
no_suspend_calls_challenge.sh	Source tree contains zero suspend / hibernate / poweroff calls	0	99 (plant a call)
scaling_horizontal_challenge.sh	5 scheduler strategies converge to consistent placements at N=10 hosts	0	99 (corrupt a strategy's score function)
stress_sustained_load_challenge.sh	Sustained load (boot/teardown) does not leak SSH sockets or fds	0	99 (skip pool close)
ui_terminal_interaction_challenge.sh	ctop TUI keystrokes drive snapshot/sort/filter end-to-end	0	99 (break the key-bind)
ux_end_to_end_flow_challenge.sh	Full multi-service boot + health + monitor + shutdown UX flow	0	99 (skip the health gate)
containers_describe_challenge.sh (round-299)	Loaded .env host list, 6 i18n bundles, governance anchors, runtime auto-detect, paired-mutation discipline	0	99 (any mutation → fail)

CONST-045 .env-driven remote-host configuration coverage

CONTAINERS_REMOTE_HOST_N_* field	Loader	Test asserting round-trip	Challenge
_NAME	pkg/envconfig/parser.go	pkg/envconfig/parser_test.go	containers_describe_challenge
_ADDRESS	same	same	same
_PORT	same	same	same

CONTAINERS_REMOTE_HOST_N_* field	Loader	Test asserting round-trip	Challenge
_USER	same	same	same
_KEY	same	same	same
_RUNTIME	same	same	same
_LABELS	same	same	same

SKIP-OK discipline (CONST-045): when CONTAINERS_REMOTE_ENABLED=false, remote-touching tests + challenges emit SKIP-OK: CONST-045 remote disabled in this environment and exit 0 (not 1) — skip is not failure, but skip is loud (per the parent \$SKIP-vs-FAIL decision tree).

CONST-050(B) compliance summary

- **Unit** — every package has *_test.go files; mocks ONLY in _test.go.
- **Integration** — tests/integration/ exercises real Docker / Podman.
- **E2E** — tests/e2e/remote_e2e_test.go against real SSH targets.
- **Security** — tests/security/ssh_security_test.go.
- **Stress** — tests/stress/distribution_stress_test.go.
- **Benchmark** — tests/benchmark/scheduler_bench_test.go.
- **Challenges** — 12 (11 pre-existing + containers_describe_challenge.sh added round-299) under challenges/scripts/.
- **Mutation** — every gate has a paired mutation in mutation_ratchet_challenge.sh; round-299 challenge ships its own paired mutation via the --mutate flag.

Anti-bluff guarantees

1. Every row's "Challenge" column points to a script that runs the real exerciser. The round-299 containers_describe_challenge.sh invokes go run ./cmd/containers-describe (when present) OR falls back to a pure-shell .env-driven describe path. Both exit 0 only on real success; mutate-mode (--mutate) deliberately corrupts a precondition and asserts exit 99.
2. CONST-045 SKIP behavior is asserted — a passing run with CONTAINERS_REMOTE_ENABLED=false MUST emit the SKIP-OK marker; absence of the marker on a skipped run is a §11.4.1 FAIL-bluff (silent skip).
3. The 6 locale bundles MUST all load without error; missing-file or parse-error on any bundle is a §11.4 PASS-bluff if the suite still reports green.

4. No metadata-only PASS — every assertion compares loaded bytes / process exit / runtime state, never just file presence.